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A SUPPORT MEMBER FOR A VEHICLE COMPONENT

Abstract

A support member for a space dividing apparatus, the support member comprising: a first support element comprising a first support strengthening rib; a second support element comprising a second support strengthening rib; and a support hinge arranged to hingeably couple the first and second support elements and having a support hinge axis, wherein the support apparatus is hingeable between: a deployed condition in which the first support strengthening rib is substantially coplanar in a first plane with the second support strengthening rib; and a stowed condition in which the first support strengthening rib is spaced from and substantially coplanar in a second plane with the second support strengthening rib, wherein the second plane is orthogonal to the first plane.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a U.S. National Phase of International Application No. PCT/EP2023/061359 entitled “A SUPPORT MEMBER FOR A VEHICLE COMPONENT,” and filed on Apr. 28, 2023. International Application No. PCT/EP2023/061359 claims priority to Great Britain Patent Application No. 2206739.1 filed on May 9, 2022. The entire contents of each of the above-listed applications are hereby incorporated by reference for all purposes.

TECHNICAL FIELD
[0002] The present disclosure relates to a support member. In particular, but not exclusively, the present disclosure relates to a support member for a vehicle component. Aspects of the invention relate to a support member and to a vehicle.
BACKGROUND
[0003] Space dividing apparatuses in vehicles and the like can often be configured in more than one position or configuration. Examples include load floors and shelves. Support members, for example gas struts, are often used to support the components in a user-desired position or configuration but known support members can use valuable packaging space and can be expensive to manufacture.
[0004] The present invention provides an improved support member.
SUMMARY OF THE INVENTION
[0005] It is an aim of the present invention to address one or more of the disadvantages associated with the prior art. Aspects and embodiments of the invention provide a support member and a vehicle as claimed in the appended claims.
[0006] Throughout the summary of the invention, description and claims the terms front, rear, upper and lower are defined with reference to their normal use in the description of a forward facing upright standing vehicle. It follows that features of the invention are described in relation to a vehicle such that, for example, front features are those closer to the front of the vehicle than rear features and upper features are those closer to the top of the vehicle than lower features.
[0007] According to an aspect of the invention there is provided a support member for a space dividing apparatus, the support member comprising: a first support element comprising a first support strengthening rib; a second support element comprising a second support strengthening rib; and a support hinge arranged to hingeably couple the first and second support elements and having a support hinge axis, wherein the support apparatus is hingeable between: a deployed condition in which the first support strengthening rib is substantially coplanar in a first plane with the second support strengthening rib; and a stowed condition in which the first support strengthening rib is spaced from and substantially coplanar in a second plane with the second support strengthening rib, wherein the second plane is orthogonal to the first plane.
[0008] Optionally, in the deployed condition, the first plane passes through (or intersects) both of the first support strengthening rib and the second support strengthening rib. The first plane may pass through (or intersect) the first and second support strengthening ribs along an entire length of

the ribs in the deployed condition. The first plane may or may not be coplanar with any surfaces of the ribs in the deployed condition.
[0009] Optionally, in the stowed condition, the second plane passes through (or intersects) both of the first support strengthening rib and the second support strengthening rib. The second plane may pass through (or intersect) the first and second support strengthening ribs along the entire length of the ribs in the stowed condition. The second plane may or may not be coplanar with any surfaces of the ribs in the stowed condition.
[0010] Optionally, the deployed position and the stowed condition are relative to a base, the second plane being parallel to a surface of the base, and the first plane extending substantially vertically from the surface of the base.
[0011] In an embodiment the first support element comprises a first surface having a recess portion; and the second support element comprises a first surface having an overlap portion arranged to locate in the recess portion of the first support element when the support apparatus is in the deployed condition.
[0012] Optionally the first and second support strengthening ribs each have a rib axis, wherein each rib axis is orthogonal to the support hinge axis.
[0013] The first support element may comprise a second surface and the first support strengthening rib may be arranged on the second surface. The second support element may comprise a second surface and the second support strengthening rib may be arranged on the second surface. In these embodiments in the stowed condition the second surface of the first support element may substantially face the second surface of the second support element.
[0014] In an embodiment, the support hinge is a barrel hinge and comprises: a first barrel; a second barrel; and a pin, wherein: the first support element comprises the first barrel; and the second support element comprises the second barrel.
[0015] Optionally the support member comprises: a first coupling hinge suitable for coupling the first support element to a first component, the first coupling hinge having a hinge axis arranged at forty-five degrees to the hinge axis of the support hinge; and a second coupling hinge suitable for coupling the second support element to a second component, the second coupling having a hinge axis arranged at forty-five degrees to the hinge axis of the support hinge, wherein: in the deployed condition the hinge axis of the first coupling hinge is substantially orthogonal to the hinge axis of the second coupling hinge; and in the stowed condition the hinge axis of the first coupling hinge is substantially parallel to the hinge axis of the second coupling hinge.
[0016] In some embodiments where the support member comprises the first component, the first component is a first coupling portion coupled to the first support element through the first coupling hinge, the first coupling portion being suitable for coupling the support member to components of a space dividing apparatus.
[0017] In some embodiments where the support member comprises the second component, the second component is suitable for coupling the support member to a partition body end of a space dividing apparatus.
[0018] Optionally the first and second coupling hinges are barrel hinges, each comprising: a first barrel; a second barrel; and a pin, wherein: the first support element comprises the first barrel of the first coupling hinge; and the second support element comprises the first barrel of the second coupling hinge.
[0019] In an embodiment the overlap portion comprises at least one overlap rib and the recess portion comprises at least one corresponding recess slot, wherein the recess slot is arranged to receive the overlap rib when the support member is in the deployed condition. The first support element may comprise a double castellation arrange to provide the recess slot and at least one stowage slot, wherein the stowage slot is arranged to receive the second support strengthening rib when the support member is in the stowed condition.
[0020] In an embodiment the support member comprises a biasing means arranged to bias the

support apparatus towards the deployed condition.
[0021] The first support may comprise a plurality of said first support strengthening rib and/or may comprise a plurality of said second support strengthening rib.
[0022] According to an aspect of the invention there is provided a vehicle comprising a support member as hereinbefore defined.
[0023] Within the scope of this application, it is expressly intended that the various aspects, embodiments, examples and alternatives set out in the preceding paragraphs, in the claims and/or in the following description and drawings, and in particular the individual features thereof, may be taken independently or in any combination. That is, all embodiments and/or features of any embodiment can be combined in any way and/or combination, unless such features are incompatible. The applicant reserves the right to change any originally filed claim or file any new claim accordingly, including the right to amend any originally filed claim to depend from and/or incorporate any feature of any other claim although not originally claimed in that manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] Embodiments of the invention will now be described, by way of example only, with reference to the accompanying drawings, in which:
[0025] FIG. **1** is a perspective view of a deployed support member in accordance with an embodiment of the invention;
[0026] FIG. **2** is an orthographic view of a stowed support member in accordance with an embodiment of the invention;
[0027] FIG. **2***a* is a cross section through the support member shown in FIG. **2**;
[0028] FIG. **3** is a perspective view of a part-deployed support member in accordance with an embodiment of the invention;
[0029] FIG. **3***a* is a section view of the support member shown in FIG. **3**;
[0030] FIG. **4** is an orthographic exploded view of a support member in accordance with an embodiment of the invention;
[0031] FIG. **5** is an orthographic view of a first support element of a support member in accordance with an embodiment of the invention;
[0032] FIG. **6** is a sectioned perspective view of a stowed support member in accordance with an embodiment of the invention;
[0033] FIG. **7** is a perspective view of a deployed space dividing apparatus in accordance with an embodiment of the invention;
[0034] FIG. **8** is a perspective view of a stowed space dividing apparatus in accordance with an embodiment of the invention;
[0035] FIG. **9** is a perspective view of a part-deployed space dividing apparatus in accordance with an embodiment of the invention;
[0036] FIGS. **10** to **13** are perspective views of sections of a space dividing apparatus in accordance with an embodiment of the invention;
[0037] FIG. **14**(*a* to d) schematically show the operation of a release button of a space dividing apparatus in accordance with an embodiment of the invention;
[0038] FIG. **15** is a perspective view of a section of a space dividing apparatus in accordance with an embodiment of the invention; and
[0039] FIG. **16** is a vehicle in accordance with an embodiment of the invention.
DETAILED DESCRIPTION
[0040] FIGS. **1** to **3** show a support member **10** in accordance with an embodiment of the invention. The support member comprises a first support element **12** and a second support element

14. In the embodiment shown, the first and second support elements are substantially triangular, or wing shaped. The first support element **12** comprises at least one first support strengthening rib **16**. The second support element **14** comprises at least one second support strengthening rib **18**. In the embodiment of FIGS. **1** to **3**, each strengthening rib **16**, **18** is a straight non-curved rib having a rib axis **20**. Each rib or rib axis is parallel to that of other ribs. In the embodiment shown all of the rib axes **20** are substantially parallel. It will be appreciated that other non-straight rib shapes are within the scope of the invention.

[0041] Each of the first support element and second support element are substantially planar components so can be described to have a first support element plane and a second support element plane respectively. It will be appreciated that this holds in embodiments where the first and second support elements do not comprise strengthening ribs.

[0042] The first and second support elements are coupled together via a support hinge **22** having a hinge axis **24** about which the support elements can rotate relative to each other. In the example embodiment, the hinge axis **24** is orthogonal to the rib axes **20**. Other relationships may be useful.

[0043] The support hinge **22** permits the support member **10** to hinge between a deployed condition **26** as shown in FIG. **1** and a stowed condition **28** as shown in FIG. **2** and described below. In the deployed condition each first support strengthening rib **16** is substantially coplanar in a first plane with each second support strengthening rib **18**. Put another way at least a portion of the ribs are substantially coplanar. The first plane is illustrated in FIG. **1** by the symbol A. In the deployed condition, axes **24**, **64** and **66** each run within the first plane, while axes **64** and **66** are substantially orthogonal.

[0044] FIG. **4** shows the deployed support member **10** of FIG. **1** in exploded form. FIG. **5** shows the first support element **12** from an opposing view to that shown in FIG. **4**. The first support element comprises a first surface **32** having a recess portion **34**. The second support element **14** comprises a first surface **33** and a corresponding overlap portion **36**. With reference back to FIG. **1**, the overlap portion **36** is arranged to locate in the recess portion **34** of the first support element **12** when the support apparatus is in the deployed condition **26**. This is advantageous in that the overlap portion prevents the support apparatus from rotating beyond the deployed condition with respect to movement from the stowed condition as will be discussed.

[0045] One or more overlap ribs **38** are provided on the overlap portion **36**. This strengthens the overlap portion and provides engagement with the recess portion as will be discussed. The presence of overlap ribs **38** is advantageous when moulding the second support element. That is to say a flat surface without ribs may present manufacturing challenges when using moulding processes.

[0046] As shown in FIG. **5**, the recess portion **34** of the first support **12** comprises recess slots **40** which, in the deployed condition **26**, accept the overlap ribs **38** of the second support element. The recess slots **40** are formed by a double castellation **42** (illustrated in FIG. **6**) which also provides stowage slots **44** which, in the stowed condition **28**, accept the strengthening ribs **18** of the second support element **14**. Specifically, the stowage slots **44** accept an unbifurcated portion of the strengthening ribs **18** as will be described.

[0047] Referring to FIG. **4**, to maximise the strength of the support member **10**, each second support strengthening rib **18** is configured, in some embodiments, to bifurcate into parallel ribs **18***a* and **18***b* at a point **46** substantially equal in distance from the support hinge **22** to the distance from the support hinge **22** to a point **46**′ where the double castellation **42** transitions into the first support element strengthening ribs **16**. This means, when stowed, the bifurcated second support strengthening ribs **18***a* and **18***b* do not collide with the double castellation **42**.

[0048] FIG. **2** shows the support member **10** in the stowed condition **28**. In this condition the first support strengthening rib **16** is shown to be spaced from and substantially coplanar in a second plane with the second support strengthening rib **18** (bifurcated as **18***a* and **18***b*). Put another way, at least a portion of the ribs are substantially coplanar. In the stowed condition **28** the rib axes are again substantially parallel (rib axes not shown in the figure as orthogonal to the plane of the figure).

[0049] It will be appreciated that during transition of the first and second support elements **12**, **14** from or to the deployed condition **26** to or from the stowed condition **28** through rotation about the hinge, the rib axes **20** of the strengthening ribs **16**, **18** of the first and second elements lose their parallelism.

[0050] The second plane is illustrated in FIG. **2** by symbol B and is orthogonal to the page showing FIG. **2**. The second plane comprises hinge axis **24**. It will be understood that relative to a fixed point outside the support member **10** and with no other rotation, the second plane B is orthogonal to the first plane A discussed in relation to the deployed condition **26**.

[0051] As mentioned, the ribs **16**, **18** are arranged on the first and second support elements so that in the stowed condition the ribs interlace. It will be appreciated that where other arrangements of ribs are used within the scope of the invention, it follows that the ribs can be configured to not interlace. However, it remains advantageous for at least a portion of the ribs to be coplanar in the stowed condition for effective collapse of the support member to a depth d**10** which is less than the sum of the depths d**12**, d**14** of each of the first and second support elements respectively. This is illustrated in FIG. **2***a*.

[0052] FIG. **3** shows the support member in a part deployed condition **30** being at a point between the stowed condition **28** and deployed condition **26**. The figure shows the aforementioned loss of parallelism of the rib axes **20** of the strengthening ribs **16**, **18** of the first and second elements in the part deployed condition **30**. It will be appreciated that in the embodiments shown in FIGS. **1** to **3**, each rib axis **20** remains orthogonal to the hinge axis **24** at all times.

[0053] FIG. **3***a* shows a section of FIG. **3**. The aforementioned transition between the double castellation **42** and the first support strengthening ribs **16** comprises a sloped region **48**. This is advantageous during manufacture of the support element to facilitate release from tooling. A corresponding shaping **50** of the overlap ribs **38** is provided to ensure maximum engagement with the double castellation **42** when in the deployed condition **26**.

[0054] FIGS. **1** to **3** also show the first support element **12** comprising a second surface **52** wherein the at least one first support strengthening rib **16** is arranged on the second surface. Similarly, the second support element **14** comprises a second surface **54** wherein the at least one second support strengthening rib **18** is arranged on the second surface. As the support member transitions from the deployed condition **26** to the stowed condition **28**, the first and second surfaces **52**, **54** come into facing each other. The provision of the strengthening ribs on the second surfaces allows for reinforcement of the first and second support elements while allowing for an effective collapse of the support apparatus to the depth d**10** as hereinbefore described.

[0055] The support hinge **22** coupling the first and second elements is a barrel hinge and comprises a first barrel **56** a second barrel **58** and a pin (not shown because the pin is inside the first and second barrels once assembled, or put another way the pin couples the barrels so is substantially contained therein). In the embodiment shown the first support element **12** comprises the first barrel **56** and the second support element **14** comprises the second barrel **58**. Other arrangements may be useful, for example in the embodiment shown, the first support element comprises two or more first barrels and the second support comprises two or more second barrels. In examples of the invention the first support element comprising the first barrel is integrally formed, for example from a single shot moulding operation. Similarly, the second support element comprising the second barrel can be integrally formed, for example from a single shot moulding operation. Other arrangements may be useful. The pin couples the first barrel to the second barrel and is coaxial with the hinge axis **24**.

[0056] In example embodiments the support member comprises a first coupling hinge **60** suitable for coupling the first support element **12** to a first component and a second coupling hinge **62** suitable for coupling the second support element **14** to a second component as will be described below. It will be understood that the couplings mentioned can be direct or indirect. The first coupling hinge has a hinge axis **64** arranged at forty-five degrees to the hinge axis **24** of the support hinge **22**. Similarly, the second coupling has a hinge axis **66** arranged at forty-five degrees to the hinge axis **24** of the support hinge **22**. The skilled person will understand that in the deployed condition the hinge axis **64** of the first coupling hinge is substantially orthogonal to the hinge axis **66** of the second coupling hinge and in the stowed condition the hinge axis **64** of the first coupling hinge **60** is substantially parallel to the hinge axis **66** of the second coupling hinge **62**.

[0057] In the embodiment shown the support member **10** comprises a first coupling portion **68** coupled to the first support element through the first coupling hinge **60**. The first coupling portion may be referred to as a base and is suitable for coupling the support member **10** to the first component as mentioned above. Similarly, the support apparatus may comprise a second coupling portion (not shown) coupled to the second support element through the second coupling hinge **62**. In some embodiments the second coupling portion is suitable for coupling the support member to the second component. It will be appreciated that within the scope of the invention, some embodiments do not comprise the first and/or second portion and instead the support member couples directly to the first and/or second component.

[0058] The first and second coupling hinges are barrel hinges, each comprising a first barrel, a second barrel and a pin. In the embodiment shown the first support element **12** comprises the first barrel **70** of the first coupling hinge **60** and the second support element **14** comprises the first barrel **72** of the second coupling hinge **62**. Each first barrel and respective support element may be integrally formed using, for example, single shot moulding as described above in relation to the support hinge.

[0059] In embodiments of the invention a support biasing means, for example a spring, is provided to generate a bias force to rotationally urge the support member **10** towards the deployed condition. The embodiment shown uses a torsion spring disposed within the support hinge **22** to urge the first and second elements **12**, **14** to rotate from each other towards the deployed condition **26**. Other arrangements may be useful. In the deployed condition **26** the support biasing means biases the support apparatus to hold it in the deployed condition. As hereinbefore mentioned, the overlap portion **36** prevents over-rotation of the support member beyond the deployed condition with respect to the stowed condition. A force greater than the bias force provided by the biasing means is therefore required to move the support apparatus from the deployed condition **26** towards the stowed condition **28**. It will be understood that the force greater than the bias force must be applied in a direction which opposes the bias force. For example, a force greater than the bias force applied in the first plane A will not result in rotation toward the stowed condition. This is useful in applications where the support member supports a space dividing apparatus because movement of items against the apparatus will not result in stowing of the support member.

[0060] FIGS. **7** to **9** show a space dividing apparatus **100** for a vehicle loadspace **102** comprising a partition body **104** and two support members **10**, **10′** as hereinbefore described. It will be understood that the two support members **10**, **10′** are mirror images of each other relative to a centre point **106** of the partition body. This permits the support members to each fold in towards the centre of the loadspace **102** as will be described. It will also be appreciated that the support members forming part of the space dividing apparatus do not comprise first support and second support strengthening ribs in all embodiments of the invention. That is to say that some embodiments feature all aspects of the above-described support apparatus excluding the first and second support strengthening ribs. The skilled person will appreciate that the below description can be applied to embodiments both comprising and excluding the first and second support strengthening ribs.

[0061] The space dividing apparatus **100** shown forms a vehicle loadspace storage solution. Other arrangements, for example, to provide a deployable shelf or table or the like are within the scope of the invention. The space dividing apparatus is arranged to move between a stowed position (FIG. **7**) and a deployed position (FIG. **8**) relative to a base **108**. In the stowed position (FIG. **8**) the one or more support members **10**, **10′** are in the stowed condition **28** and in the deployed position (FIG.

7) the one or more support members **10**, **10′** are in the deployed condition **26**. FIG. **9** illustrates a part deployed position of the space dividing apparatus in which the support members **10**, **10′** are in the part deployed condition **30**.

[0062] In the deployed position, the space dividing apparatus provides a storage volume **110** defined as the volume within the partition body **104** and the two support members **10**, **10′**. This advantageously divides the vehicle loadspace **102** so that items which fit within the storage volume **110** can be retained and held closer to the loadspace entry (typically known as the tailgate of the vehicle) for easy removal. The partition body **104** can be used in the deployed configuration to provide flexible storage solutions **112** that are sometimes but not always wanted by users of the vehicle. In the example where the space dividing apparatus provides a shelf or table or the like, the space dividing apparatus is installed orthogonally to arrangement shown in FIG. **7**. Here in the deployed configuration, the partition body provides the shelf or table or the like. It will be appreciated that items could be suspended from the shelf or table or the like in the storage volume which, in this example configuration, lies underneath the partition body.

[0063] The first surface **32** of the first support element and the first surface **33** of the second support element are arranged to face the centre of the loadspace **102**. Advantageously the first surfaces are smooth, i.e. do not comprise ribs, and thus provide a surface that does not snag items placed in the storage volume **110**. Furthermore, the positioning of the support member means that the support hinge **22** faces away from the centre of the storage volume. This reduces the likelihood that items will catch on the support hinge. The arrangement also allows the support member **10** to collapse towards the centre of the loadspace **102** meaning that the support member can be placed at or near to ends **114** of the partition body **104**, or put another way at the extremity of the loadspace, to maximise storage area.

[0064] In some embodiments, a load floor unit **116** comprises the base **108**. A load floor section **118** is provided adjacent the base **108**. In some embodiments the base and the load floor section are integral, thus advantageously reducing part count to provide a floor for the loadspace **102**. In other embodiments the base **108** may be attached to the load floor section. In either configuration it will be understood that the load floor unit **116** is removable for users of the vehicle to access sub-floor areas where for example a spare wheel for the vehicle may be stored. In examples of the invention, reversible snap fittings are used to fix the base and/or the load floor section to a scuff plate and/or body in white of the vehicle. This allows the user to remove and reattach the space dividing apparatus from and to the vehicle. Examples of the reversible snap fittings include ball clips. Other fittings may be useful.

[0065] Removal of the space dividing apparatus is, in some examples, facilitated by one or more base straps **119** shown in FIG. **10**. The base strap is optionally fixed to the base **108** using, for example, an adhesive. In embodiments the base strap is not fixed to the base but remains proximal thereto. FIG. **10** shows the base strap with the first coupling portion **68** removed. In this embodiment the base strap is held through compression of the first coupling portion **68** against the base **108**. FIG. **11** shows the underside of the first coupling portion. Fixing of the first coupling portion to the base retains the base strap **119**. The base strap passes through a strap aperture **69** (shown in FIG. **1**) in the first coupling portion **68**.

[0066] FIG. **10** also illustrates an embodiment in which the space dividing apparatus comprises a base cover **109**. The base cover comprises, in examples of the invention, a non-slip material to aid the retention of items stored in the storage volume. In one example the base cover **109** is made from rubber. Other materials may be useful. The fixing of the first coupling portion to the base **108** acts to clamp the base cover in place. In some examples the space dividing apparatus comprises a base lip **107** proximal to the loadspace opening in use. The base lip advantageously acts to create a tray with the first coupling portion and load floor section. This can be a benefit to, for example, retain liquids if they are spilled in the storage volume. In an example, the base lip is made from a material that has some flexibility so that the partition body slightly compresses the base lip in the stowed position. This acts to prevent the partition body from vibrating in the stowed position and creating unwanted noise.

[0067] As shown in FIGS. **7** to **9**, the partition body **104** comprises a board, which may be formed from an extrusion or the like. The partition body has a material surface **120** and a function surface **122**. The material surface can comprise a carpet, for example, to match the adjacent load floor section **118**. The function surface is presented to a user of the space dividing apparatus when in the deployed position. In some embodiments of the invention, the function surface comprises elasticated straps **112** arranged to advantageously retain goods positioned against the function surface. One or more elastic strap tensioner **113** is provided to section the elasticated strap. In some embodiments each strap tensioner **113** is moveable across the function surface to advantageously permit a user to tailor the available elasticated strap size, for example to retain either a flask, a football, a frisbee, a spade, a tent, and the like.

[0068] It will be understood that throughout the following description, reference to the support member is applicable to either or both of the support members **10**, **10**′ forming part of the space dividing apparatus. For ease of description, however, only one support member in the space dividing apparatus is described.

[0069] Ends **114** of the partition body **104** can be integrally formed with or attached to the partition body. In the example shown the ends **114** are formed separate to the partition body and then fixed thereto. With reference to FIG. **12**, each end **114** comprises second barrels **124** arranged to couple with the first barrel **72** of the second element **14** of the support member **10** to form the second coupling hinge **62**. In embodiments of the invention, the ends **114** comprise a track **126** in which a release button **130** is slidably housed. The release button is used to aid the user in overcoming the bias force of the support member.

[0070] Shown in FIG. **13**, the release button **130** comprises a first button ramp **132** having a ramp surface **134** configured to engage with a ramp surface **136** of a support ramp **138** provided as part of examples of the support member **10**. Application of force to the release button in the direction indicated by arrow **140** causes the ramp surfaces **134** and **136** to engage. As the release button is held in the track **126**, the engagement causes the support member **10**, or more specifically the second element **14** of the support member, to rotate away from the deployed condition and from the release button as the bias force is overcome. This results in movement of the support member from the deployed condition. In some embodiments of the invention, continued or further depression of the button results in the ramp surfaces being overcome. Movement back towards the deployed condition is promoted by the bias force but is in some embodiments prevented by ramp edges **142**, **144** which become proximal after the ramp surfaces are overcome. The support member is thus held in a part deployed condition. This means that force applied in the A plane (reference made to FIG. **1** and supporting description) results in a component of that force acting against the bias force thus leading to further rotation of the support member toward the stowed condition. The movement of the release button and the effects thereof is described further with reference to schematic FIGS. **14***a* to **14***d*.

[0071] As described in the foregoing, the release button is configured to initiate movement of the support member from the deployed condition towards the stowed condition. This is shown schematically in FIG. **14** in which, four stages of are shown in subfigures a) to d).

[0072] In an example embodiment, the release button **130** is movable along a button axis **150**. The button is biased towards a presented condition by a button bias provided by, for example, a button spring **152**. When the space dividing apparatus is in the deployed position, the button ramp **132** is aligned with and opposes the support ramp **138**, as shown in FIG. **14***a*. As will be explained, movement or depression of the button **130** along the button axis **150** causes the ramp surface **134** of the button ramp to engage with the ramp surface **136** of the support ramp and urges the second support element away from the button axis **150** and away from the deployed condition.

[0073] FIG. **14***a* shows the button **130** and a portion of the second support element **14** when the space dividing apparatus is in the deployed position as mentioned above. The button is in a presented condition. The ramp surface **134** of the button ramp **132** faces or is in contact with the ramp surface **136** of the support ramp **138**. As the button is depressed (FIG. **14***a* through **14***b*), for example by a user using their thumb, in the direction indicated by arrow **140**, the ramp surfaces **134**, **136** engage one another and the second support element **14** is urged away, shown by arrow **152**, from the button **130**. This has the effect of overcoming the bias force provided by the biasing means of the support member and enables movement of the space dividing apparatus from the deployed condition to the stowed condition.

[0074] Further depression of the button (FIG. **14***b* through FIG. **14***c*) results in the ramp surfaces **134**, **136** passing each other and the support element **14** returning towards the button **130** under the bias force provided by the biasing means as described above. As shown in FIG. **14***c*, return of the button through the button bias is prevented by each ramp **132**, **138** ramp edge **142**, **144** which engage. In some embodiments of the invention, the button is configured so that a portion of the button above the ramp edge **142** is wider than a portion of the button below the ramp **132**. In other embodiments, such as the one shown in FIG. **13**, the second support element **14** is configured so that a portion of the support element is wider (or closer to the button) below the support ramp **138** than it is above the support ramp. In either configuration, this means that once the ramp surfaces **134**, **136** have passed each other, the second support member returns, via the biasing means, to a position that is offset from the deployed condition. This is exemplified in FIG. **14***c* where the deployed condition is shown by dashed line **154** and the offset position is shown by dashed line **156**. Advantageously this prevents the space dividing apparatus from returning to the fully deployed position. In use this permits a user to press the button to initiate a move from the deployed position towards the stowed position before then applying force to the partition body to continue to overcome the biasing means and move the space dividing apparatus to the stowed position.

[0075] FIG. **14***d* schematically shows the arrangement of the button **130** and the portion of the second support element **14** when the space dividing apparatus is in the stowed position according to an embodiment of the invention. As can be seen in the figure by way of comparison to FIGS. **14***a* to **14***c*, the second support element can be considered to have rotated away from the button. That is to say, the support ramp **138** of the support element **14** no longer faces or opposes the button ramp **132** of the button **130**. As the ramp edges **142**, **144** no longer engage, the button returns to its presented configuration. In the embodiment schematically shown, the button is prevented from extending beyond the presented configuration by a catch **158** catching a ledge **160** provided on the second support element. The catch is biased against the ledge by the button bias **152**.

[0076] It will be appreciated that the rotation away of the support element from the button is caused by the arrangement of the space dividing apparatus which sees the second support element rotate about the second coupling hinge as the apparatus transitions between the deployed and stowed conditions. In turn it will be understood that as the space dividing apparatus is moved from the stowed position to the deployed position, the button and portion of the second support element return to their aligned condition as shown in FIG. **14***a*. It follows that the ledge **160** positioned on the second support element **14** is presented to the button **130** both when the support ramp **138** faces and does not face the button ramp **132**.

[0077] FIGS. **14***a* and **14***d* show the top of the button aligning with the top of the second support element. However, in some embodiments the button can be presented above the second support element. Other arrangements may be useful.

[0078] In some embodiments of the invention, the release button is prevented from extending beyond the presented configuration by a boss **162** positioned in the track **126**, as shown in FIG. **12**. A channel **164** in the button **130** receives the boss and a dead stop at each end of the channel prevents the button from extending beyond the presented configuration and advantageously from being depressed further than is useful to overcome the bias force.

[0079] Returning to FIG. **7**, the base **108** comprises a raised portion **166**. The raised portion is shaped to allow the support member to stow beside it when the space dividing apparatus is in the stowed position. This is illustrated through FIG. **9** which shows the support members **10**, **10′** stowing beside the raised portion **166**. In some embodiments, the raised portion has a depth equal to that of the stowed support member (d**10** as discussed above). In some embodiments there is a small gap formed between the raised portion **166** and the partition body **104** when the space dividing apparatus is in the stowed position. This is advantageous in allowing for manufacturing tolerances. Support for heavy objects placed on the material surface is provided by the partition body bearing on the base at the ends **114** of the partition body. In some embodiments the raised portion provides support to the partition body when the space dividing apparatus is in the stowed position. This means that heavy objects placed on the material surface do not result in a flex of the surface.

[0080] In an example embodiment the space dividing apparatus comprises a pull loop **168**, attached to the partition body **104**. The pull loop is provided to permit a user to urge the space dividing apparatus from the stowed position by lifting the partition body **104**. In some embodiments the space dividing apparatus is releasably retained in stowed position by gravity acting on the partition body. In other embodiments a latching means is used to retain the space dividing apparatus in the stowed position.

[0081] In examples comprising the latching means for retention of the space dividing apparatus in the stowed position, the release button **130** is used to releasably latch the partition body to the base. FIG. **15** shows the space dividing apparatus in the stowed position with the end **114** of the partition body (as shown in other figures) removed to enable a view of the latching means. In an example, the button comprises a button latching ramp **170** configured to engage with a base latching ramp **172** in the stowed position. The latching ramps are configured so that when a user pulls on the pull loop, the ramps act to depress the button against the button bias **152**. Sufficient force applied to the pull loop causes the ramps to be overcome and the space dividing apparatus to move to a part-deployed position. It follows that when stowing the space dividing apparatus, the button **130** must be depressed to overcome the latching ramps in a reverse direction to that achieved by the user pulling the pull loop. In some embodiments the latching ramps are double sided so that application of a downward force to stow the space dividing apparatus similarly acts against the button bias **152**. In either example, the button bias advantageously acts to retain the space dividing apparatus in the closed position.

[0082] The first coupling portion **68**, in some embodiments, comprises the base latching ramp **172**. In such embodiments the first coupling portion is attached to the base **108**. In other example embodiments, the base **108** comprises the base latching ramp. Other arrangements may also be useful.

[0083] It will be appreciated in the foregoing that some embodiments of the invention comprise a pair of release buttons, each positioned at the ends of the partition body and configured to engage with the respectively positioned support members. In use, a user having a deployed space dividing apparatus can choose to simultaneously press the button of both release buttons or can press one at a time before then moving the partition body to return the space dividing apparatus to the stowed condition. The button and support ramp edges as hereinbefore described prevent the support apparatuses from returning to a deployed configuration. Advantageously this can support single handed operation of the space dividing apparatus.

[0084] FIG. **16** schematically shows a vehicle **200** comprising a space dividing apparatus or a support apparatus as hereinbefore described.

[0085] It will be appreciated that various changes and modifications can be made to the present disclosed examples without departing from the scope of the present application as defined by the appended claims. Whilst endeavouring in the foregoing specification to draw attention to those features believed to be of particular importance it should be understood that the Applicant claims protection in respect of any patentable feature or combination of features hereinbefore referred to and/or shown in the drawings whether or not particular emphasis has been placed thereon.

Claims

1. A support member for a space dividing apparatus, the support member comprising: a first support element comprising a first support strengthening rib; a second support element comprising a second support strengthening rib; and a support hinge arranged to hingeably couple the first and second support elements and having a support hinge axis, wherein the support member is hingeable between: a deployed condition in which the first support strengthening rib is substantially coplanar in a first plane with the second support strengthening rib; and a stowed condition in which the first support strengthening rib is spaced from and substantially coplanar in a second plane with the second support strengthening rib, wherein the second plane is orthogonal to the first plane.
2. The support member as claimed in claim 1, wherein: the first support element comprises a first surface having a recess portion; and the second support element comprises a first surface having an overlap portion arranged to locate in the recess portion of the first support element when the support member is in the deployed condition.
3. The support member as claimed in claim 1, wherein the first and second support strengthening ribs each have a rib axis, wherein each rib axis is orthogonal to the support hinge axis.
4. The support member as claimed in claim 1, wherein: the first support element comprises a second surface and the first support strengthening rib is arranged on the second surface; and the second support element comprises a second surface and the second support strengthening rib is arranged on the second surface, wherein in the stowed condition the second surface of the first support element substantially faces the second surface of the second support element.
5. The support member as claimed in claim 1, wherein the support hinge is a barrel hinge and comprises: a first barrel; a second barrel; and a pin, wherein: the first support element comprises the first barrel; and the second support element comprises the second barrel.
6. The support member as claimed in any of the preceding claim 1, comprising: a first coupling hinge suitable for coupling the first support element to a first component, the first coupling hinge having a hinge axis arranged at forty-five degrees to the hinge axis of the support hinge; and a second coupling hinge suitable for coupling the second support element to a second component, the second coupling hinge having a hinge axis arranged at forty-five degrees to the hinge axis of the support hinge, wherein: in the deployed condition the hinge axis of the first coupling hinge is substantially orthogonal to the hinge axis of the second coupling hinge; and in the stowed condition the hinge axis of the first coupling hinge is substantially parallel to the hinge axis of the second coupling hinge.
7. The support member as claimed in claim 6, comprising the first component, the first component being a first coupling portion coupled to the first support element through the first coupling hinge, the first coupling portion being suitable for coupling the support member to components of the space dividing apparatus.
8. The support member as claimed in claim 6, comprising the second component, the second component being suitable for coupling the support member to a partition body end of the space dividing apparatus.
9. The support member as claimed in any of claim 6, wherein the first and second coupling hinges are barrel hinges, each comprising: a first barrel; a second barrel; and a pin, wherein: the first support element comprises the first barrel of the first coupling hinge; and the second support element comprises the first barrel of the second coupling hinge.
10. The support member as claimed in claim 2, wherein the overlap portion comprises at least one overlap rib and the recess portion comprises at least one corresponding recess slot, wherein the at least one corresponding recess slot is arranged to receive the at least one overlap rib when the

support member is in the deployed condition.

11. The support member as claimed in claim 10 wherein the first support element comprises a double castellation arrange to provide the at least one corresponding recess slot and at least one stowage slot, wherein the at least one stowage slot is arranged to receive the second support strengthening rib when the support member is in the stowed condition.

12. The support member as claimed in claim 1 comprising a biasing means arranged to bias the support member towards the deployed condition.

13. The support member as claimed in claim 1 wherein the first support element comprises a plurality of said first support strengthening rib.

14. The support member as claimed in claim 1 wherein the second support element comprises a plurality of said second support strengthening rib.

15. A vehicle comprising the support member as claimed in claim 1.